

Mutual Fund Analytics - Project Report

Prepared by: R SHYAAM KRISHNAN

Project: Bluestock Fintech Mutual Fund Analytics Capstone

1. Project Overview

The Mutual Fund Analytics project analyzes Indian mutual fund datasets to generate actionable insights for investors and analysts. Python was used for data ingestion, cleaning and analytics, SQLite for database management, SQL for querying, and Power BI for dashboard development.

2. Objectives

- Clean and validate mutual fund datasets.
- Analyze NAV, AUM, SIP inflows, investor behaviour and fund performance.
- Build an interactive Power BI dashboard.
- Generate business insights for data-driven investment decisions.

3. Data Sources

Ten datasets were used: Fund Master, NAV History, AUM by Fund House, Monthly SIP Inflows, Category Inflows, Industry Folio Count, Scheme Performance, Investor Transactions, Portfolio Holdings and Benchmark Indices. Live NAV data was also fetched using MFAPI.

4. Methodology

Data ingestion → Data cleaning → Validation → SQLite database → SQL analytics → Exploratory Data Analysis → Performance Analytics → Power BI Dashboard.

5. Exploratory Data Analysis

Performed NAV trend analysis, AUM growth analysis, SIP trend analysis, category inflow analysis, investor demographic analysis, benchmark comparison and portfolio analysis.

6. Performance Analytics

Evaluated 1-Year, 3-Year and 5-Year returns, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Expense Ratio and Morningstar Ratings to compare fund performance.

7. Dashboard Development

The Power BI dashboard consists of Executive Overview, Fund Performance, Investor Analytics and Market Analytics pages with interactive slicers and KPIs.

8. Key Findings

- NAV values show consistent long-term growth.
- SIP inflows increased steadily, indicating higher investor participation.
- Equity-oriented funds attracted the largest investments.
- Large fund houses dominated Assets Under Management.
- Tier-1 cities contributed the highest number of investors.
- Benchmark indices reflected sustained market growth.
- Risk-adjusted metrics helped identify high-performing mutual fund schemes.

9. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, SQLite, SQL, Power BI, Git and GitHub.

10. Conclusion

The project demonstrates a complete end-to-end data analytics workflow from raw data processing to interactive dashboard creation. The developed solution provides meaningful insights into mutual fund performance and investor behaviour, enabling informed financial decision-making.